

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-5: Image Enhancement in the Spatial Domain

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-3)

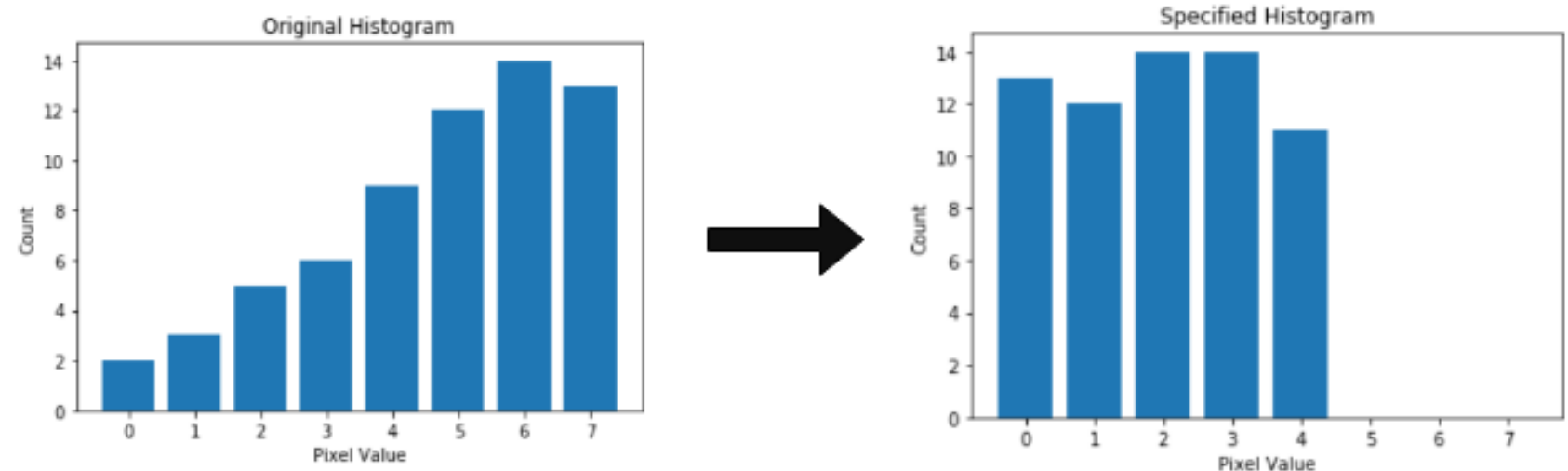
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4.4 Histogram Matching or Histogram Specification

Histogram Matching or Histogram Specification is a technique used to enhance the contrast of an image, similar to histogram equalization.

However, instead of transforming the image to have a uniform histogram (as in histogram equalization), this method transforms the image so that its histogram matches a specified shape.

This gives more control over the enhancement process.



Example

Input Image Intensities: [0,1,2,3,4]

Histogram of the image: $h_r = [4,4,6,4,2]$

Target Histogram: $h_z = [3,3,4,6,4]$

Step 1: Compute CDFs

Input Image CDF:

$$T_r(r) = \frac{\text{cumulative sum of } h_r}{\text{total pixels}}$$

$$T_r = \left[\frac{4}{20}, \frac{8}{20}, \frac{14}{20}, \frac{18}{20}, \frac{20}{20} \right] = [0.2, 0.4, 0.7, 0.9, 1.0]$$

r=0, r=1, r=2, r=3, r=4

Target Histogram CDF:

$$T_z(z) = \frac{\text{cumulative sum of } h_z}{\text{total pixels}}$$

$$T_z = \left[\frac{3}{20}, \frac{6}{20}, \frac{10}{20}, \frac{16}{20}, \frac{20}{20} \right] = [0.15, 0.3, 0.5, 0.8, 1.0]$$

$z=0, z=1, z=2, z=3, z=4$

Step 2: Mapping

For each $T_r(r)$, find the closest $T_z(z)$ and map r to z :

$$T_r = \left[\frac{4}{20}, \frac{8}{20}, \frac{14}{20}, \frac{18}{20}, \frac{20}{20} \right] \stackrel{r=0, r=1, r=2, r=3, r=4}{=} [0.2, 0.4, 0.7, 0.9, 1.0]$$

$$T_z = \left[\frac{3}{20}, \frac{6}{20}, \frac{10}{20}, \frac{16}{20}, \frac{20}{20} \right] \stackrel{z=0, z=1, z=2, z=3, z=4}{=} [0.15, 0.3, 0.5, 0.8, 1.0]$$

$$T_r(0) = 0.2 \rightarrow \text{Closest } T_z = 0.3 \rightarrow \text{Map } r = 0 \text{ to } z = 1$$

$$T_r(1) = 0.4 \rightarrow \text{Closest } T_z = 0.5 \rightarrow \text{Map } r = 1 \text{ to } z = 2$$

$$T_r(2) = 0.7 \rightarrow \text{Closest } T_z = 0.8 \rightarrow \text{Map } r = 2 \text{ to } z = 3$$

$$T_r(3) = 0.9 \rightarrow \text{Closest } T_z = 1.0 \rightarrow \text{Map } r = 3 \text{ to } z = 4$$

$$T_r(4) = 1.0 \rightarrow \text{Closest } T_z = 1.0 \rightarrow \text{Map } r = 4 \text{ to } z = 4$$

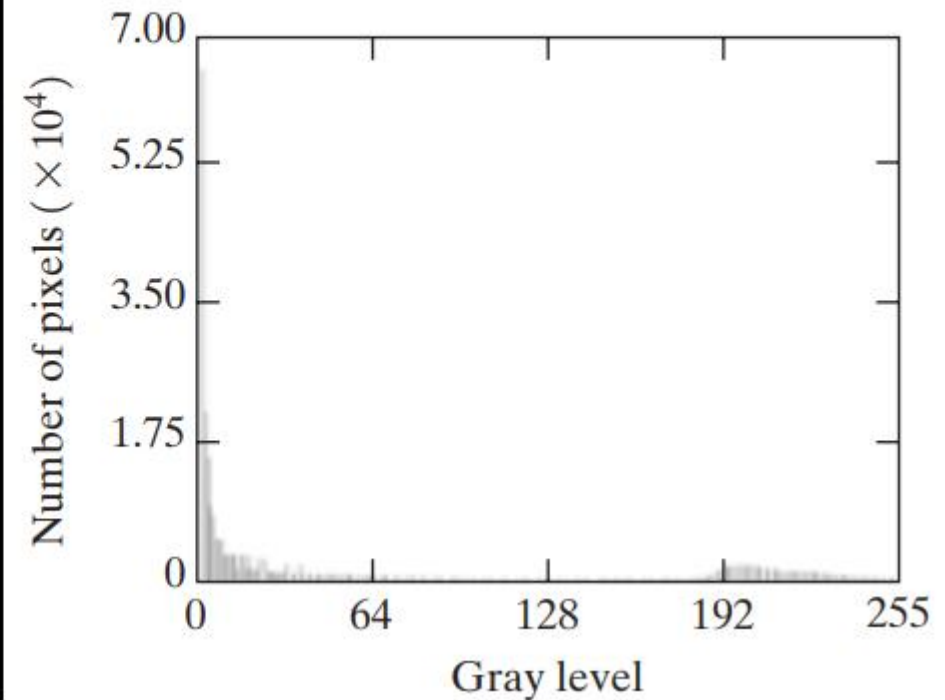
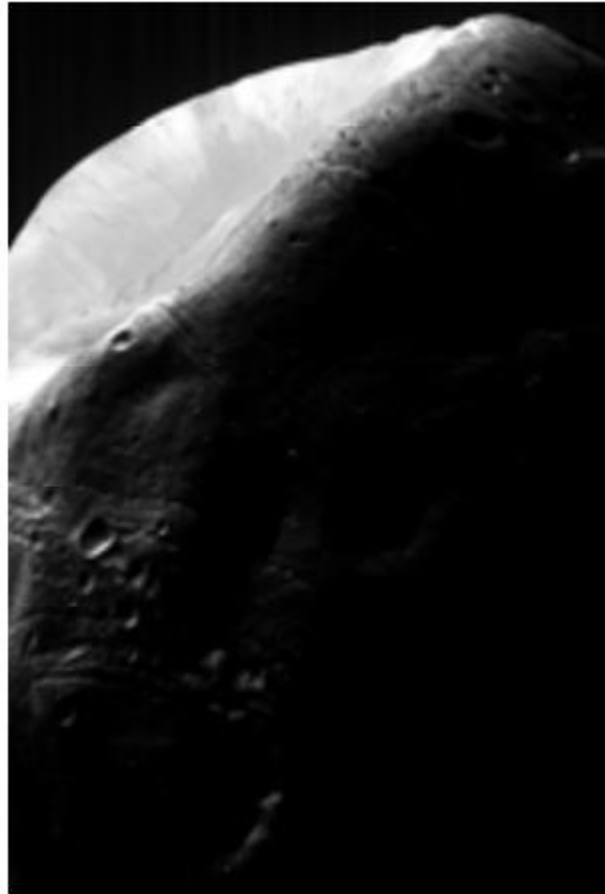
Step 3: Output Image

After mapping, the input intensities $[0, 1, 2, 3, 4]$ become $[1, 2, 3, 4, 4]$.

4.5 Comparison between histogram equalization and histogram matching

Figure shows an image of the Mars moon, Phobos, taken by NASA's Mars Global Surveyor.

The image is dominated by large, dark areas, resulting in a histogram characterized by a large concentration of pixels in the dark end of the gray scale.



At first glance, one might conclude that histogram equalization would be a good approach to enhance this image, so that details in the dark areas become more visible.

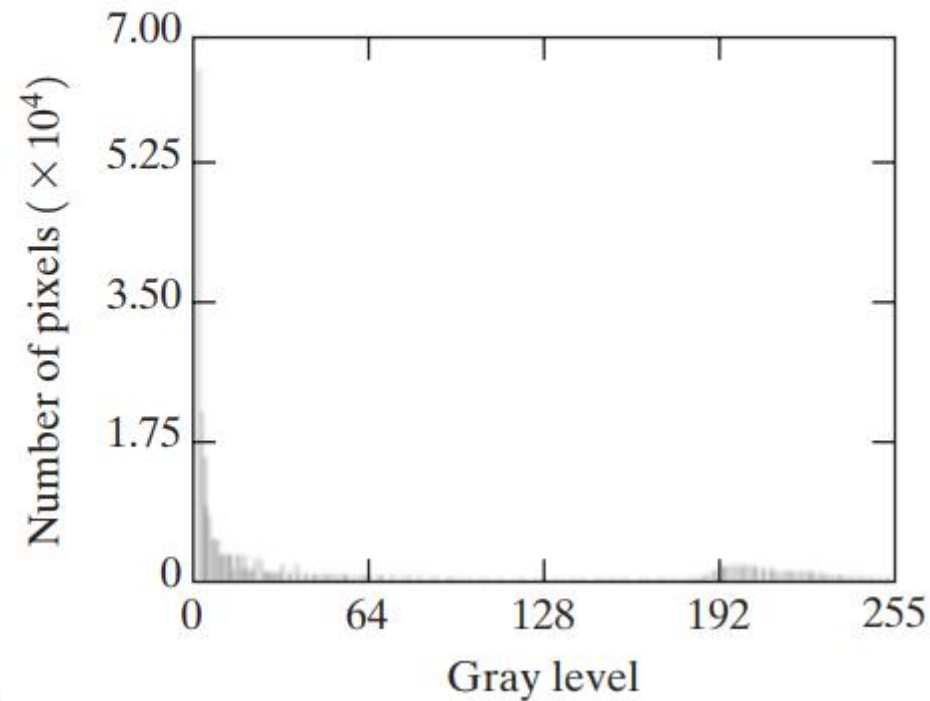
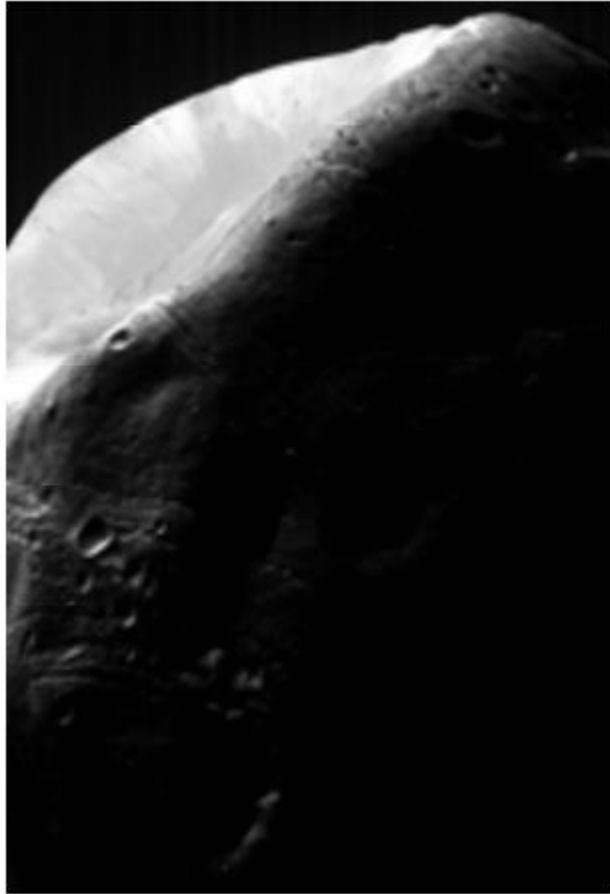
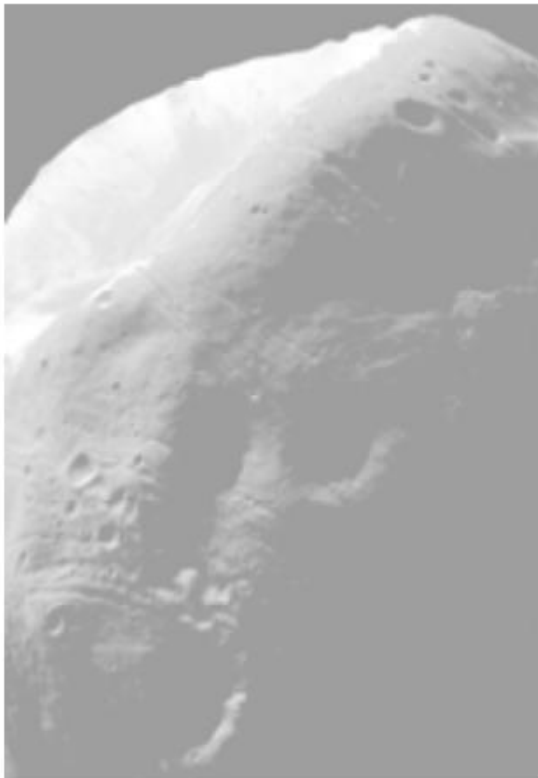
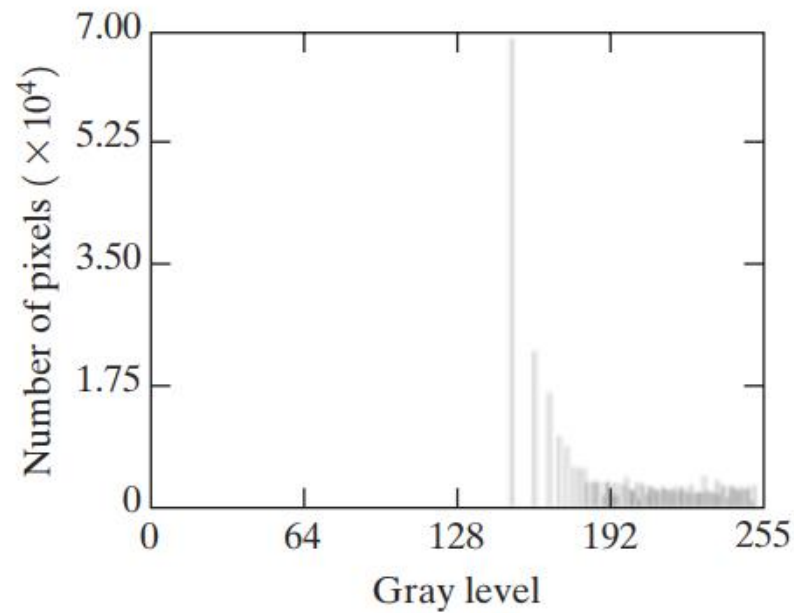
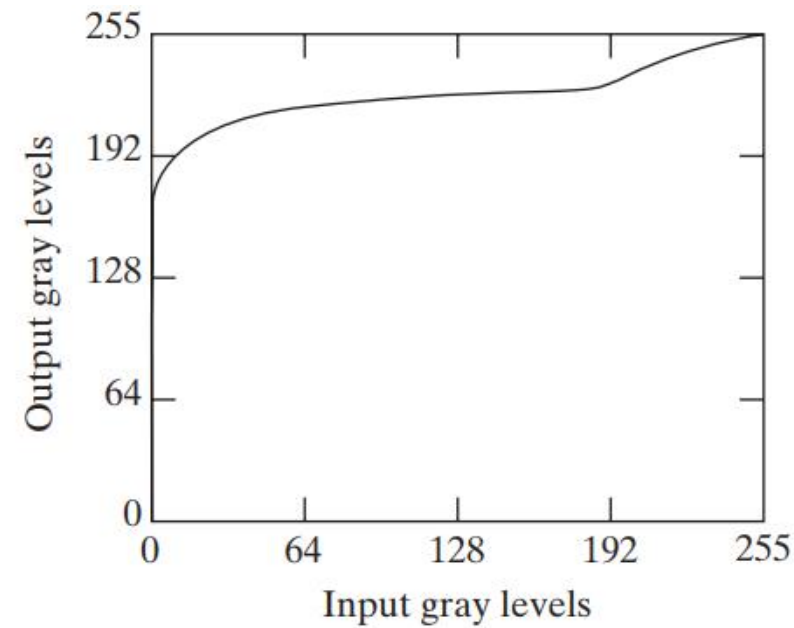


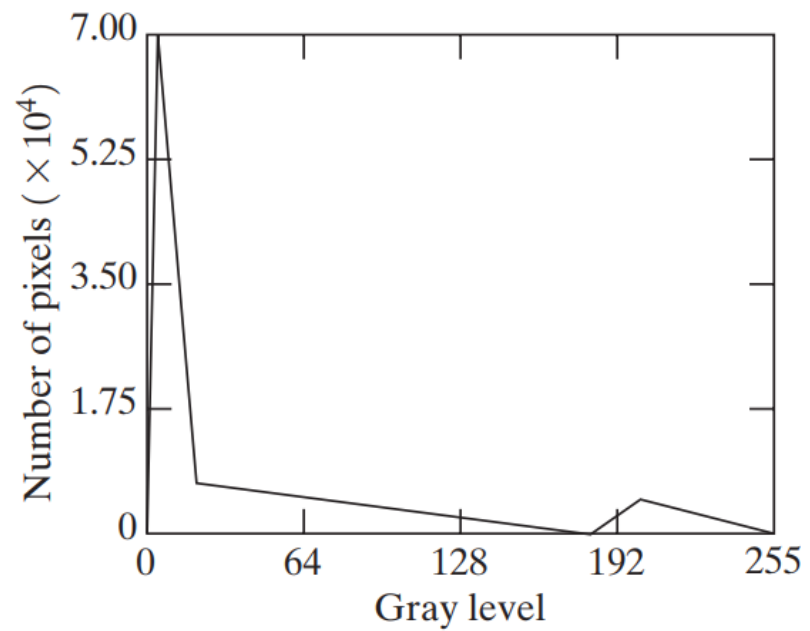
Figure shows the histogram equalization transformation

Transformation
function for
histogram
equalization

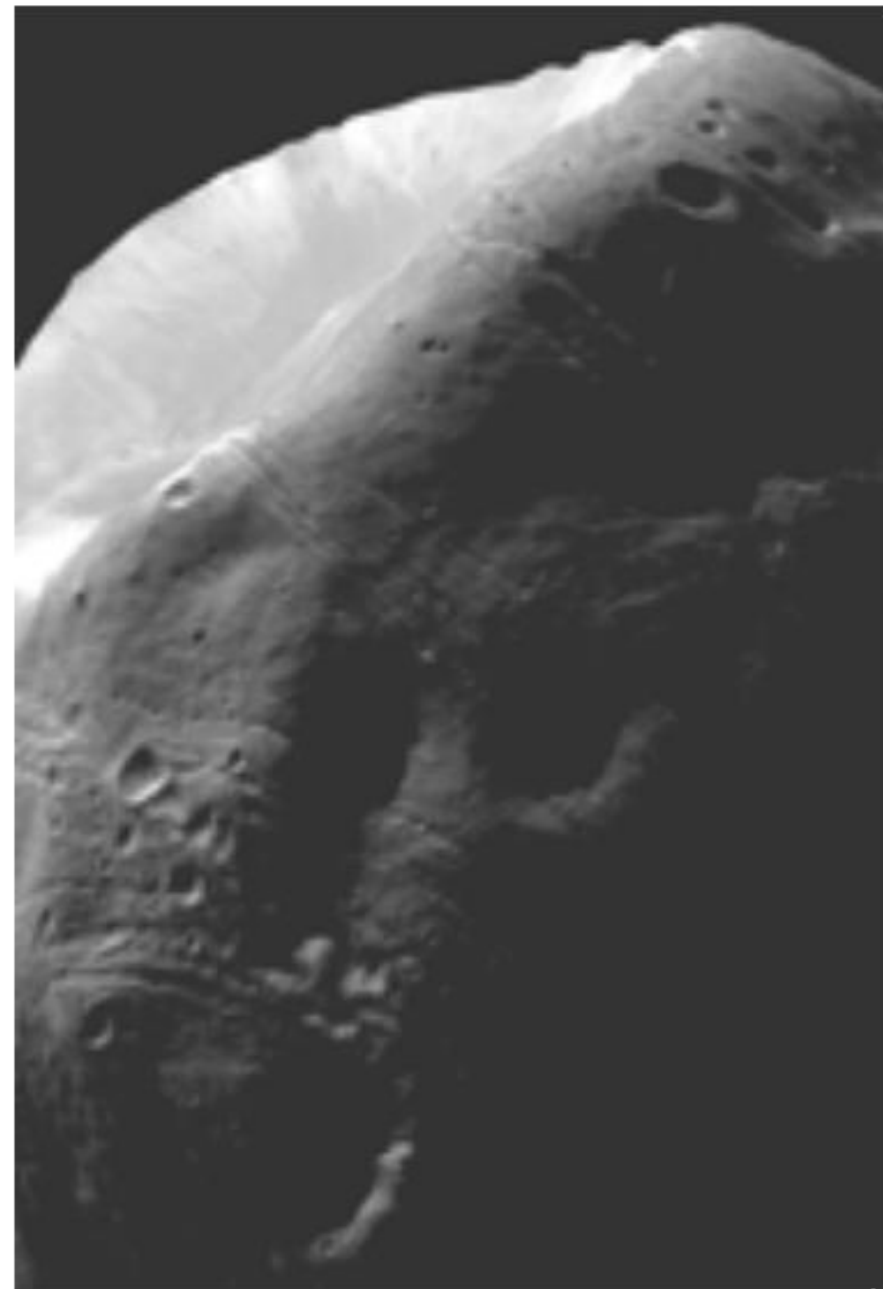
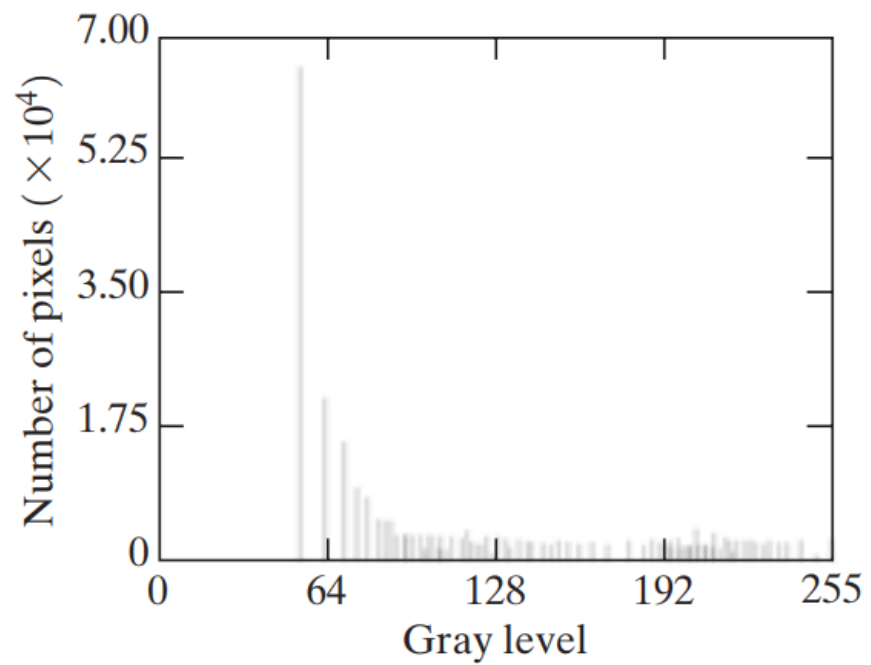


Histogram equalized image

Specified
histogram



Histogram



5. Local Enhancement

Global Methods

The histogram processing methods are global, in the sense that pixels are modified by a transformation function based on the **gray-level content of an entire image**.

Concept behind local methods

Although this global approach is suitable for overall enhancement, there are cases in which it is necessary to **enhance details over small areas** in an image.

Why not Global methods?

The number of pixels in these areas may have **negligible influence** on the computation of a global transformation whose shape does not necessarily guarantee the desired local enhancement.

How to guarantee the desired local enhancement?

The solution is to devise transformation functions based on the gray-level distribution—or other properties—in the **neighborhood of every pixel** in the image.

How to adapt the histogram processing in local enhancement?

The histogram processing techniques are easily adaptable to local enhancement. The procedure is to **define a square or rectangular neighborhood** and move the center of this area from pixel to pixel.

At each location, the histogram of the points in the neighborhood is computed and either a **histogram equalization or histogram specification** transformation function is obtained.

This function is finally used to map the gray level of the pixel centered in the neighborhood. The center of the neighborhood region is then **moved to an adjacent pixel location** and the **procedure is repeated**.

Figure (a) shows an image that has been slightly blurred to reduce its noise content.

Figure (a)

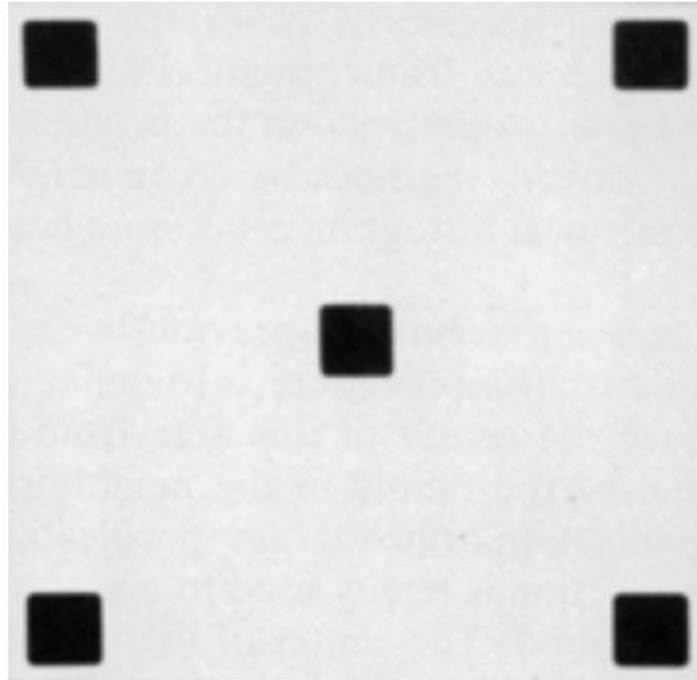


Figure (b)

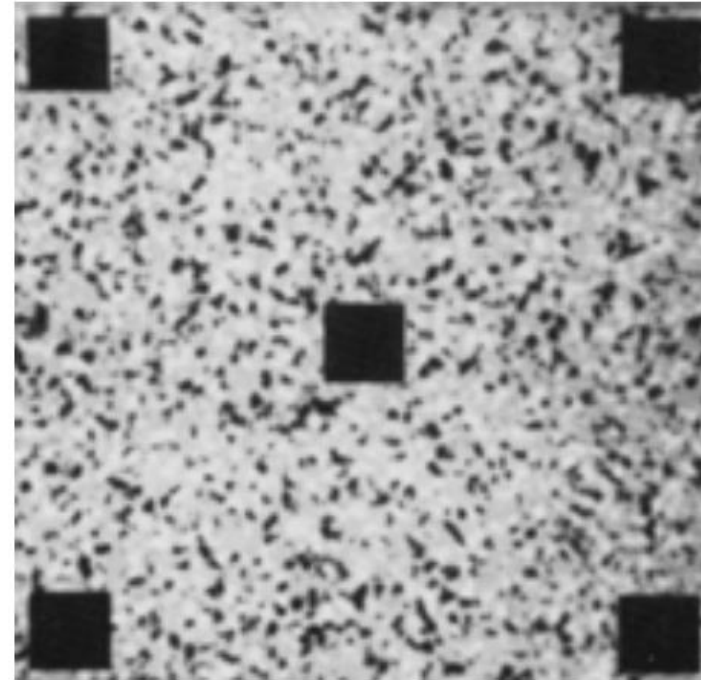


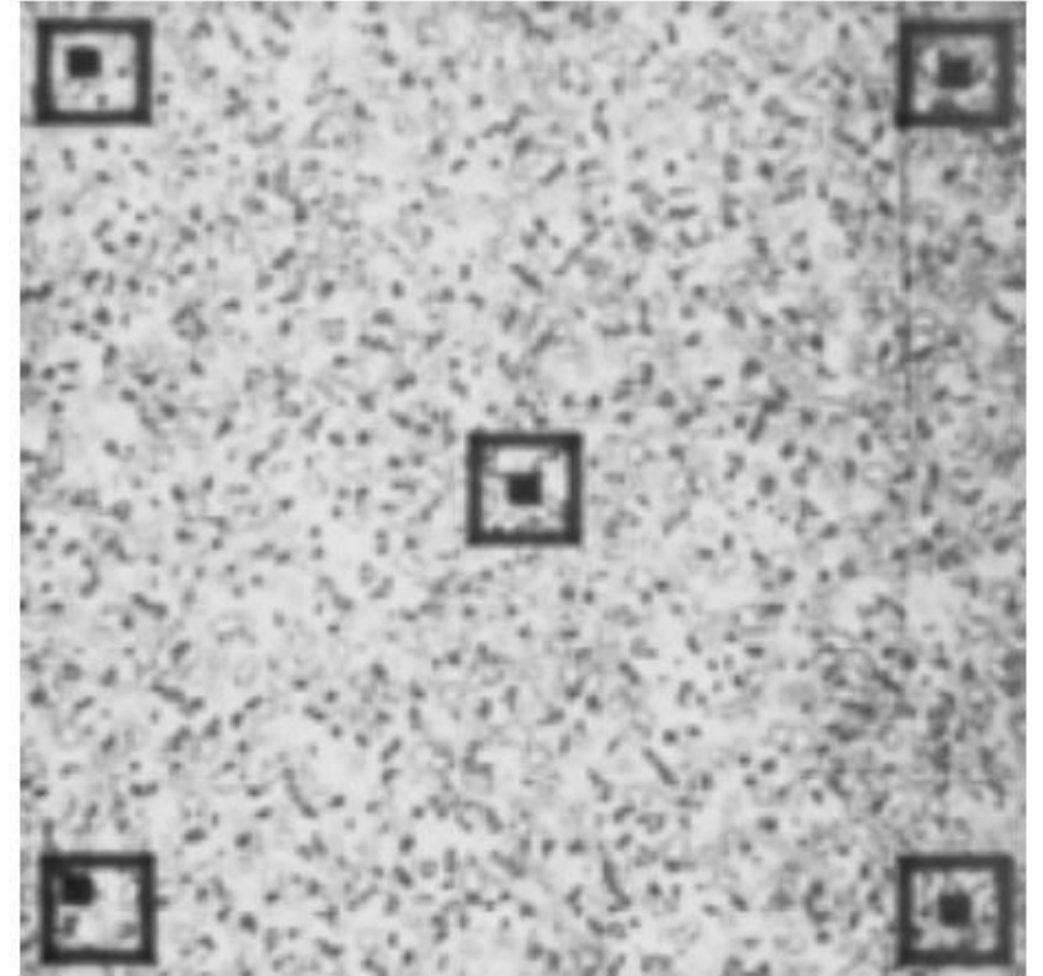
Figure (b) shows the result of global histogram equalization.

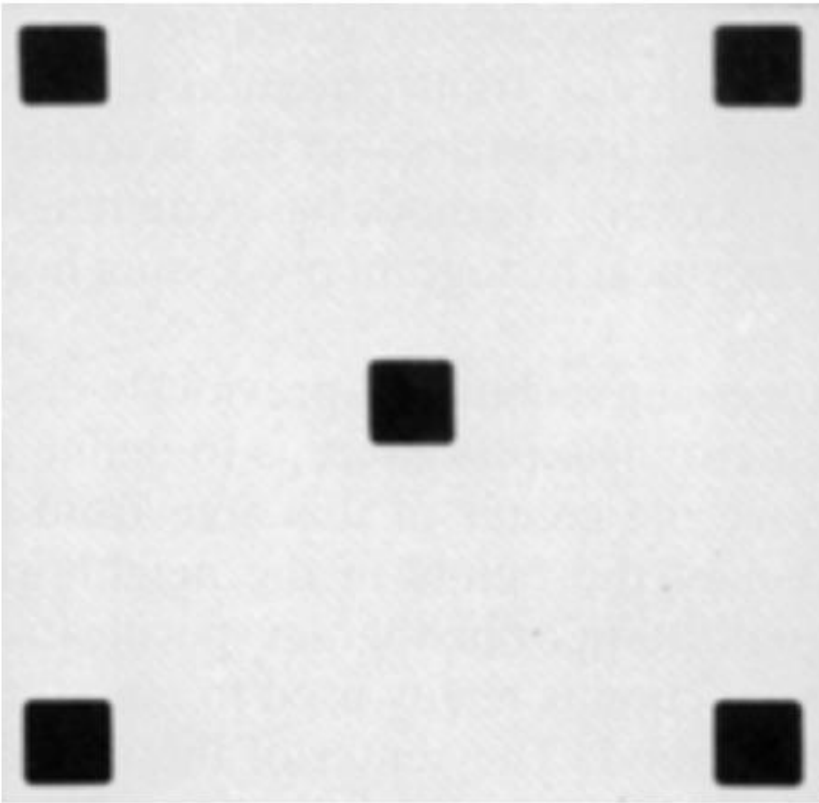
As is often the case when this technique is applied to smooth, noisy areas, Fig. (b) shows considerable enhancement of the noise, with a slight increase in contrast. Note that **no new structural details were brought** out by this method.

However, local histogram equalization using a 7 X 7 neighborhood revealed the presence of small squares inside the larger dark squares.

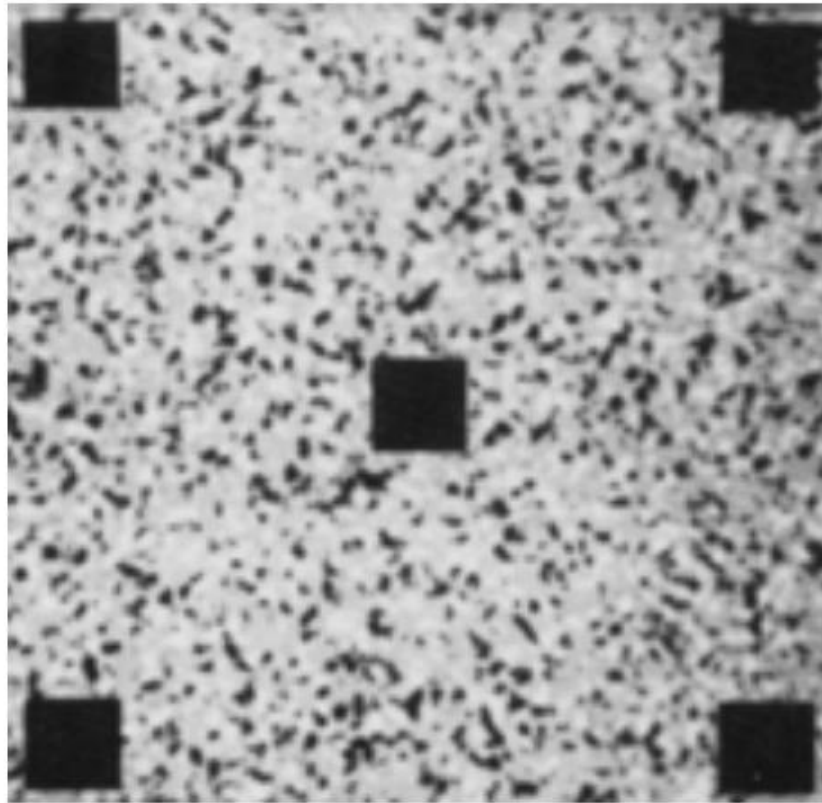
The small squares were too close in gray level to the larger ones, and their sizes were too small to influence global histogram equalization significantly.

Note also the finer noise texture is a result of local processing using relatively small neighborhoods.

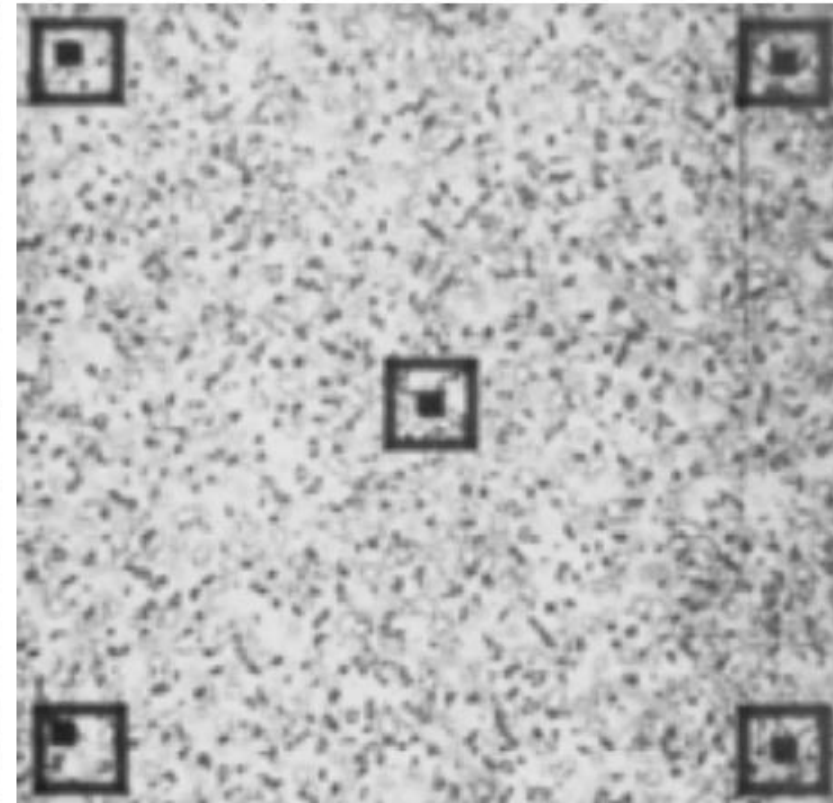




(a) Original image



(b) Result of global
histogram equalization



(c) Result of local histogram
equalization using a 7 X 7
neighborhood about each
pixel.

6. Enhancement Using Arithmetic/Logic Operations

The **NOT logic** operator performs the negative transformation.

The **AND and OR operations** are used for masking; that is, for selecting subimages in an image.

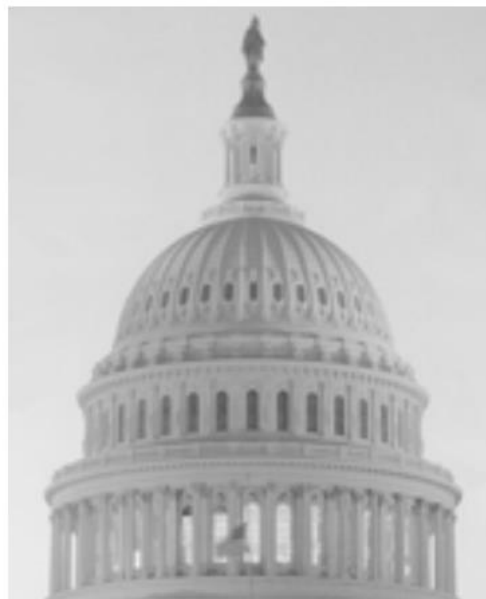
In the AND and OR image masks, light represents a binary 1 and dark represents a binary 0.

Masking sometimes is referred to as region of interest (ROI) processing.

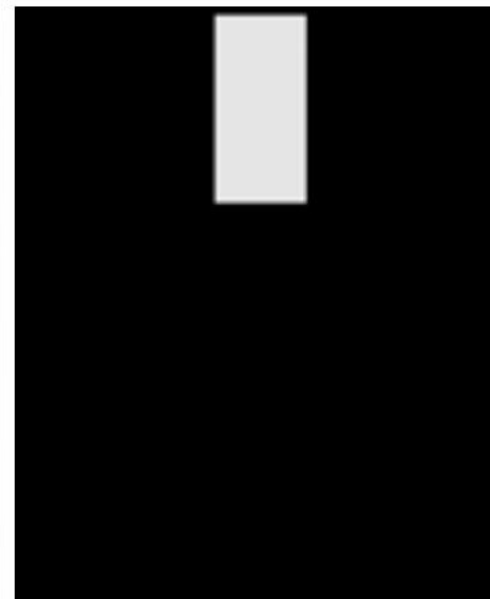
❑ In terms of enhancement, masking is used primarily to isolate an area for processing. This is done to highlight that area and differentiate it from the rest of the image.

$$\text{AND}(p_1, p_2) = p_1 \wedge p_2$$

Original image



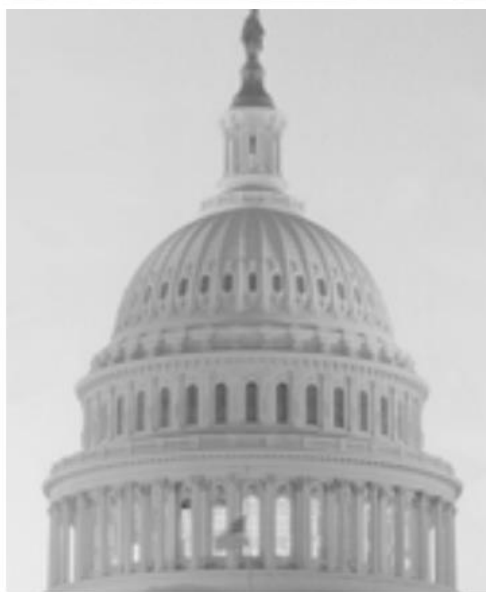
AND image mask



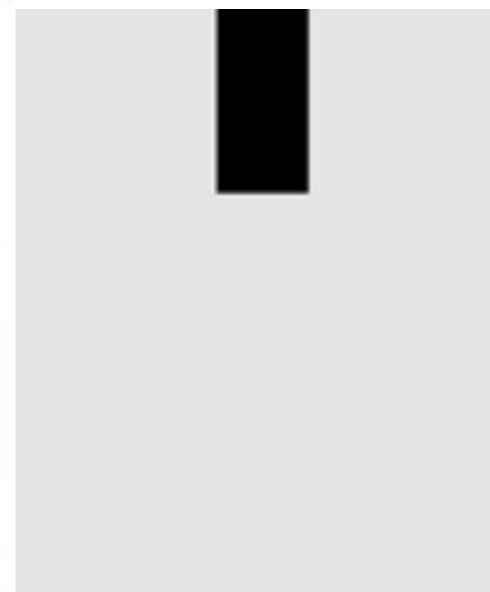
Result of AND operation



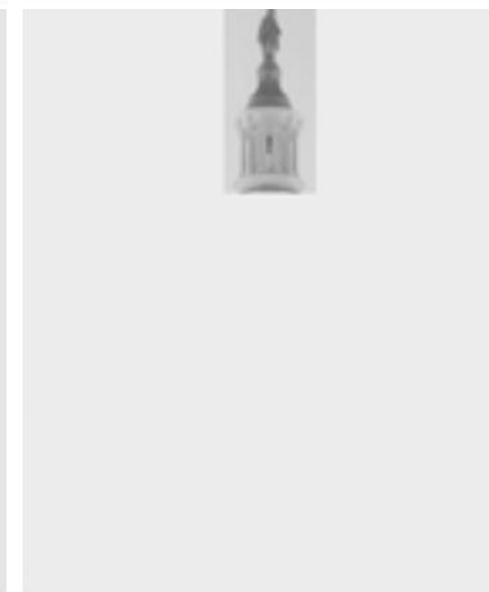
$$\text{OR}(p_1, p_2) = p_1 \vee p_2$$



Original image



OR image mask



Result of OR operation

6.1 Image Subtraction

The difference between two images $f(x, y)$ and $h(x, y)$, expressed as

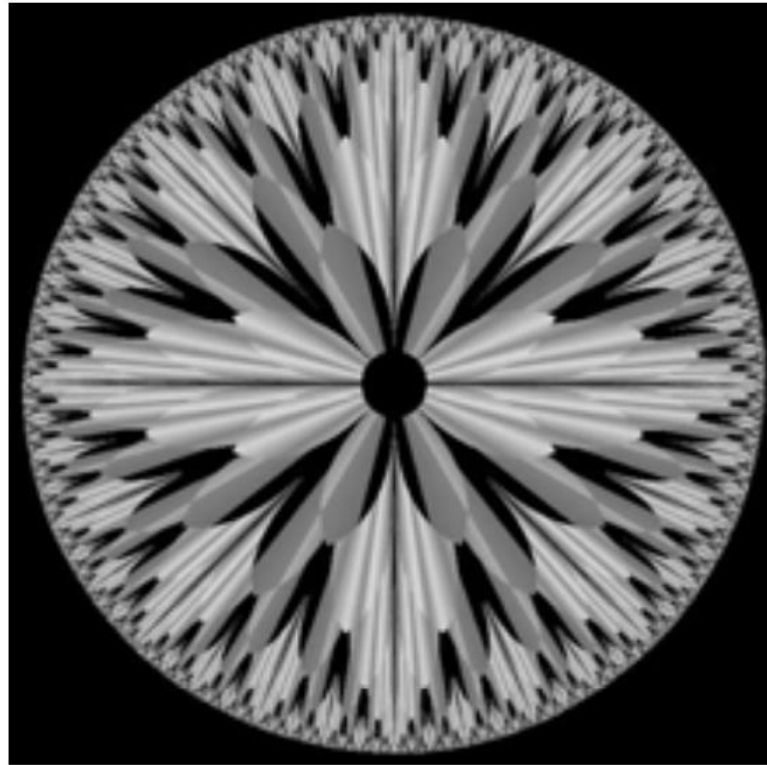
$$g(x, y) = f(x, y) - h(x, y)$$

is obtained by computing the difference between all pairs of corresponding pixels from f and h .

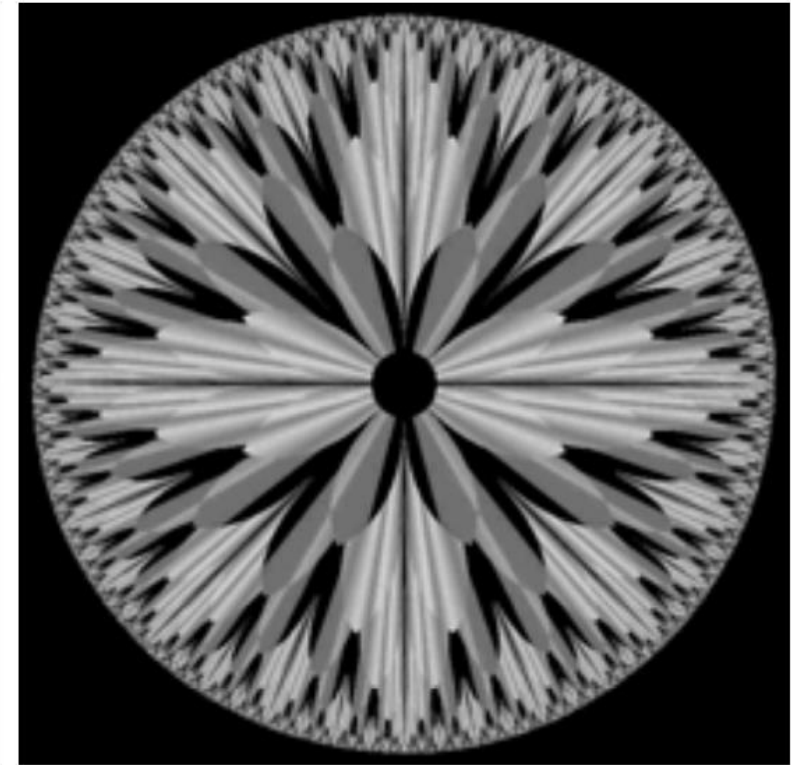
The key usefulness of subtraction is the enhancement of differences between images.

Figure (b) shows the result of discarding (setting to zero) the four least significant bit planes of the original image.

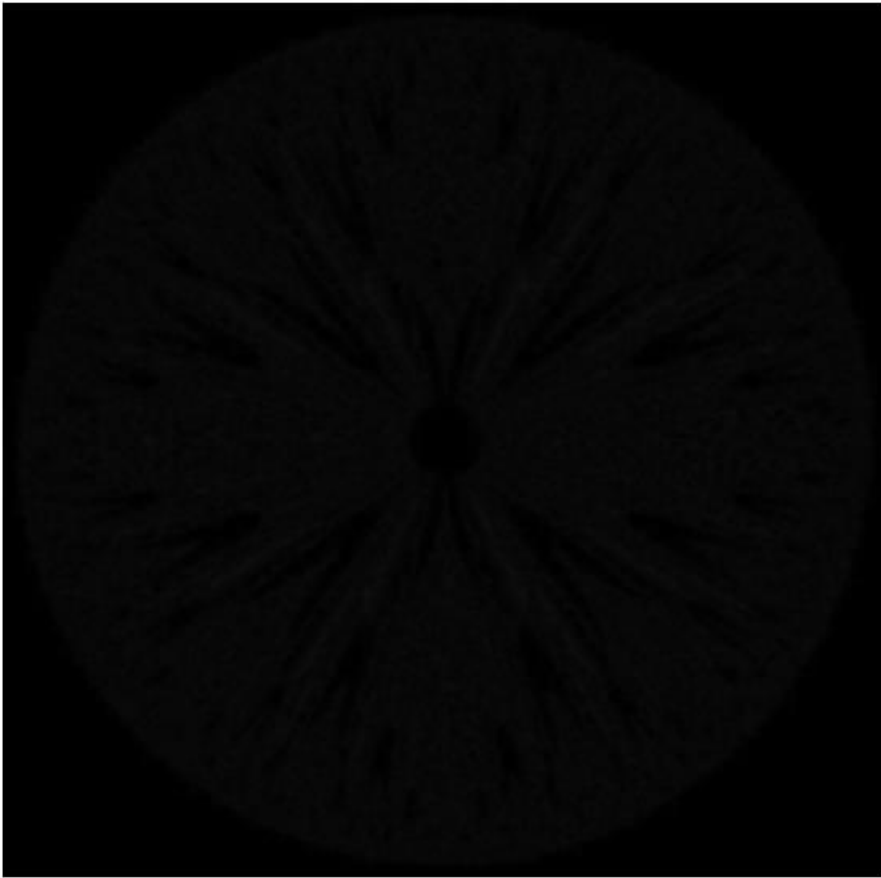
The images are nearly identical visually, with the exception of a very slight drop in overall contrast due to less variability of the gray level values in the image.



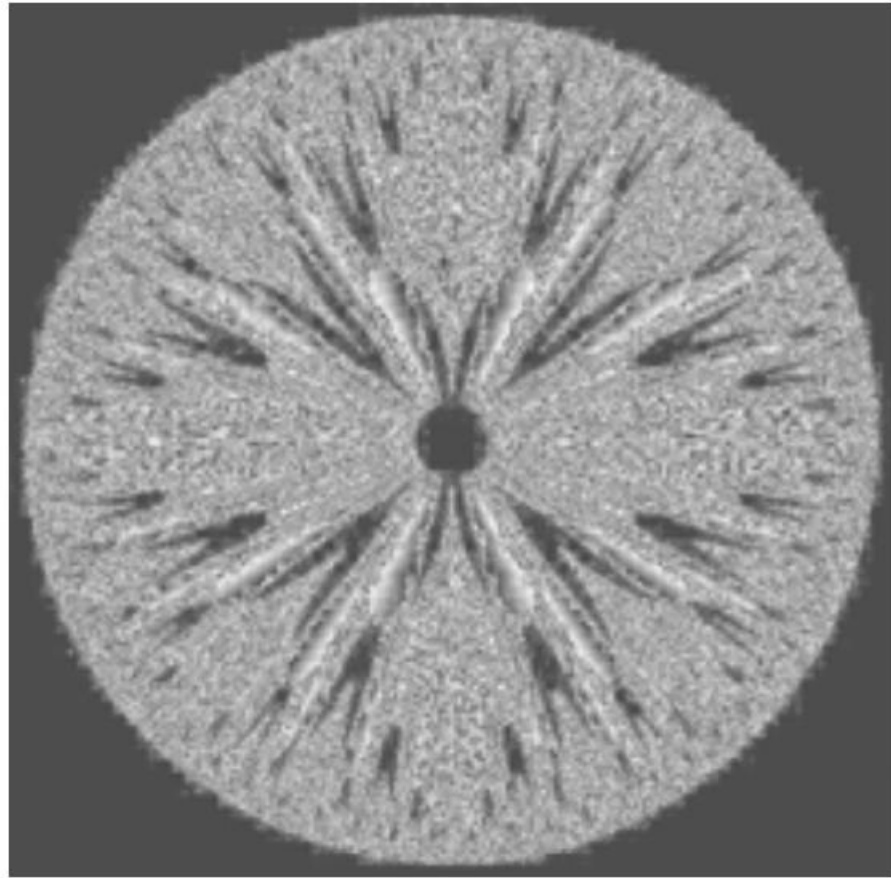
a) original fractal image



b) Result of setting the four lower-order bit planes to zero.



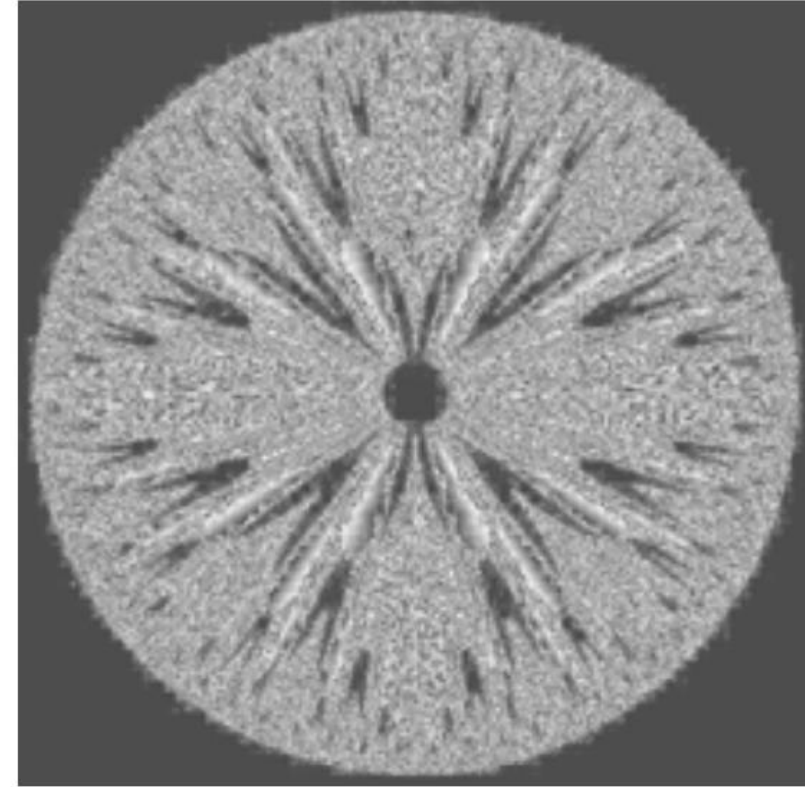
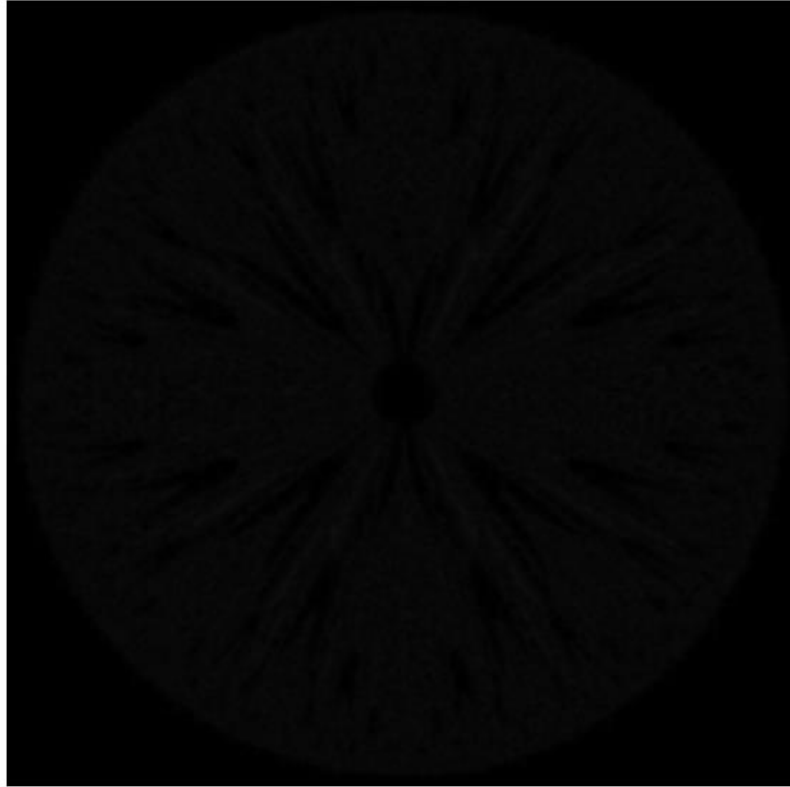
c) Difference between (a) and (b)



d) Histogram equalized difference image

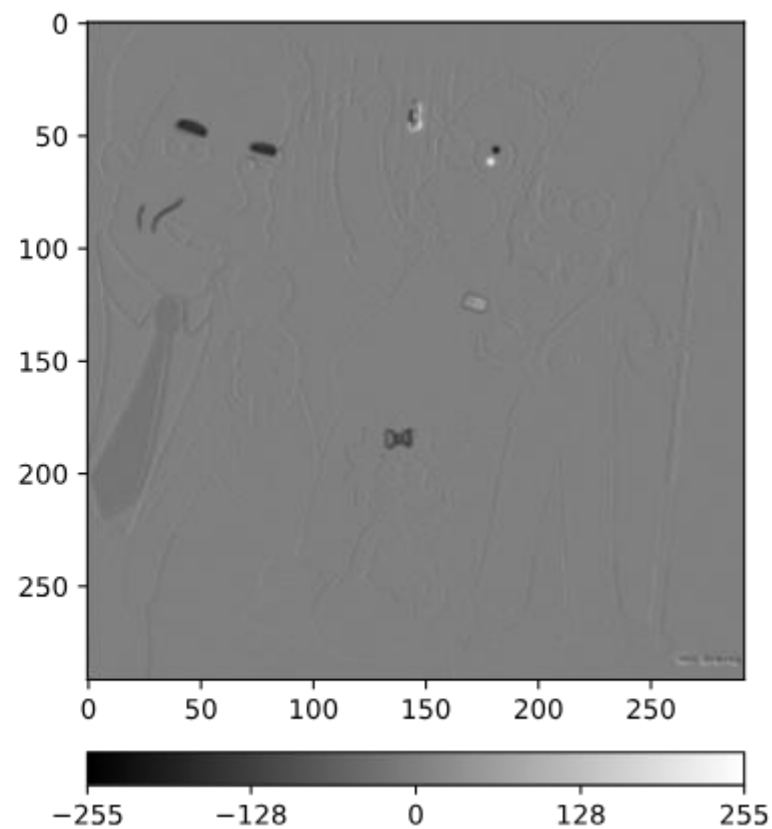
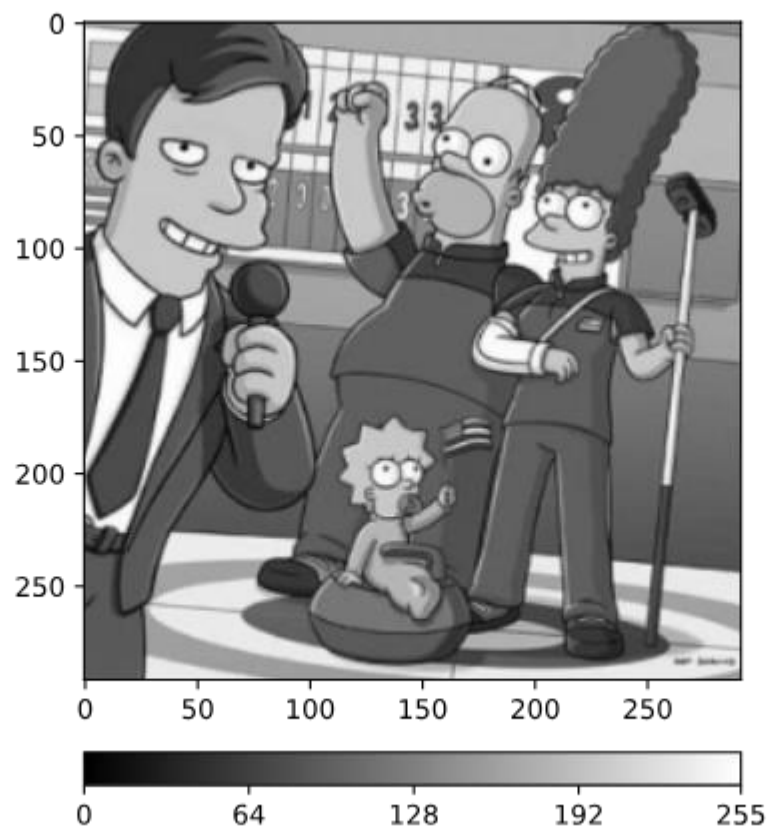
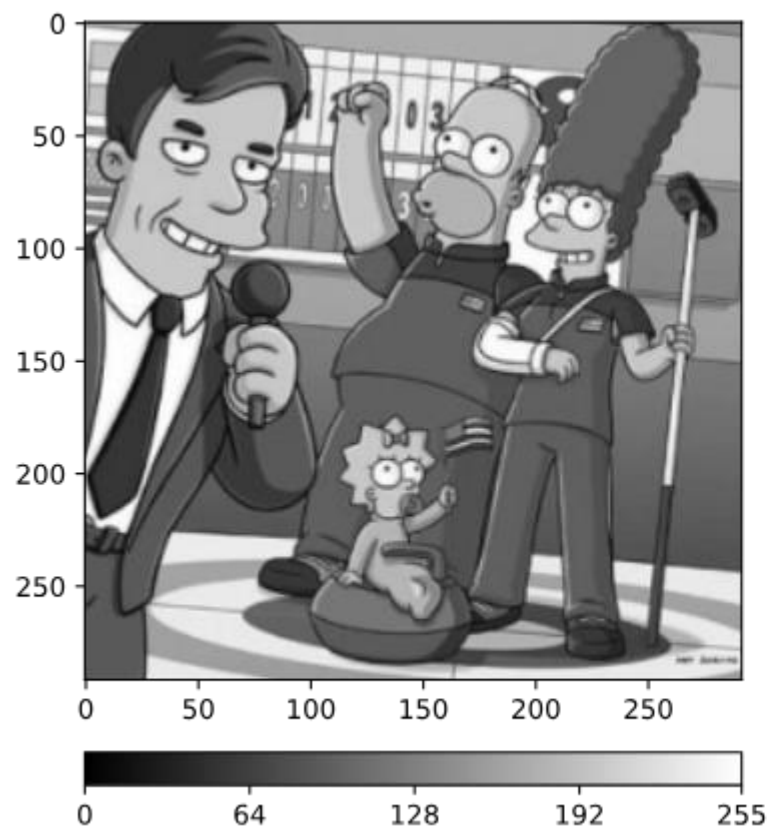
The differences in pixel values are so small that the difference image appears nearly black when displayed on an 8-bit display.

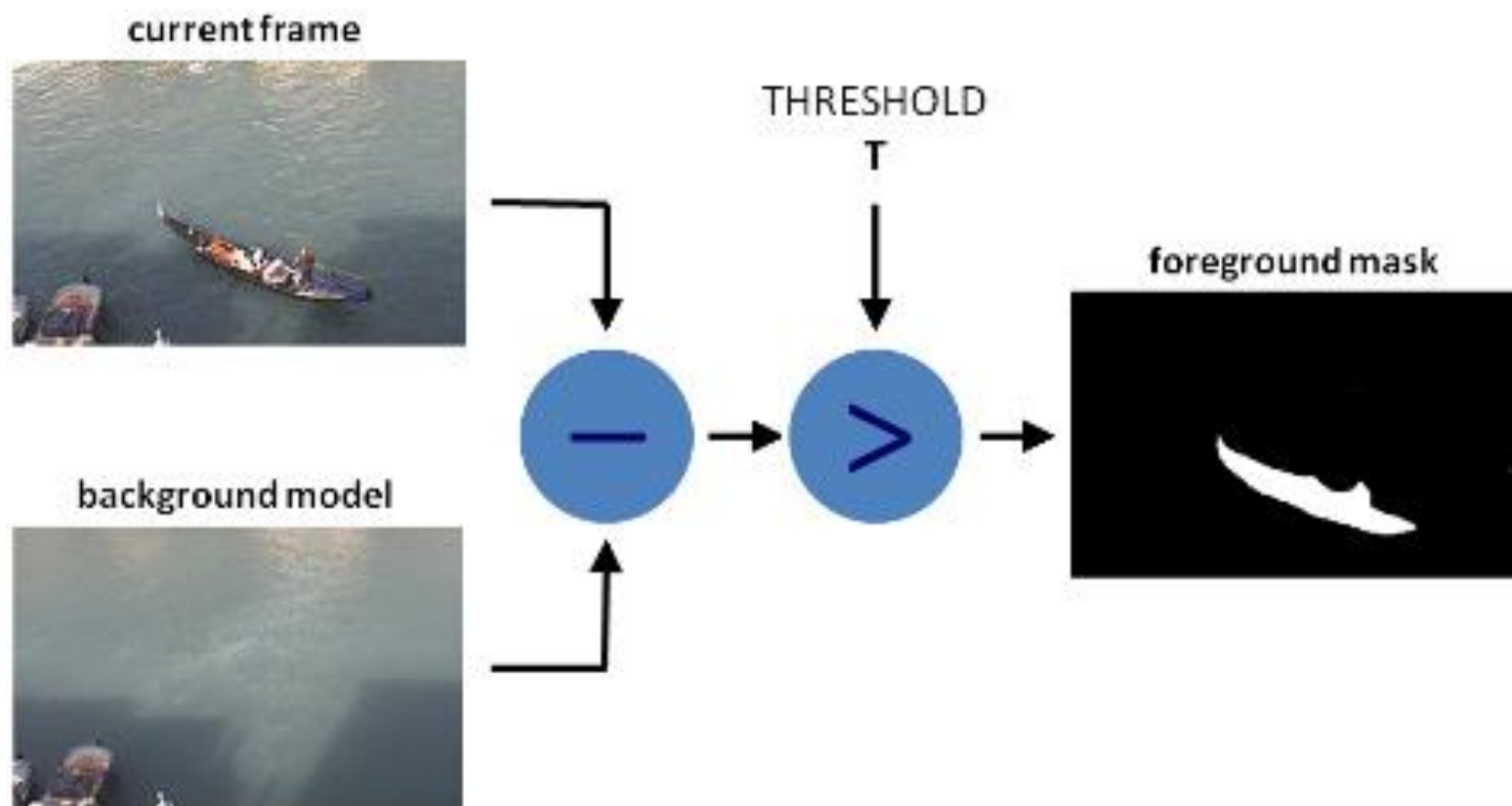
In order to bring out more detail, we can perform a contrast stretching transformation.



c) Difference between (a) and (b) d) Histogram equalized difference image

We chose histogram equalization, but an appropriate power-law transformation would have done the job also.





6.2 Image Averaging

Consider a noisy image $g(x, y)$ formed by the addition of noise $\eta(x, y)$ to an original image $f(x, y)$; that is,

$$g(x, y) = f(x, y) + \eta(x, y)$$

where the assumption is that at every pair of coordinates (x, y) the noise is uncorrelated and has zero average value.

- ✓ The objective of the following procedure is to reduce the noise content by adding a set of noisy images, $\{g_i(x, y)\}$.

if an image $\bar{g}(x, y)$ is formed by averaging K different noisy images

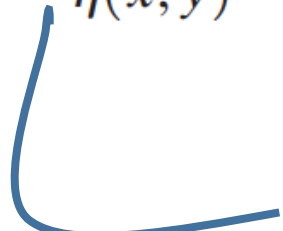
$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

then it follows that all at coordinates (x, y)

The expected value of $\bar{g}$

$$E\{\bar{g}(x, y)\} = f(x, y)$$

The variances of $\bar{g}$ $\sigma_{\bar{g}}^2(x, y) = \frac{1}{K} \sigma_{\eta(x, y)}^2$

The variances of η

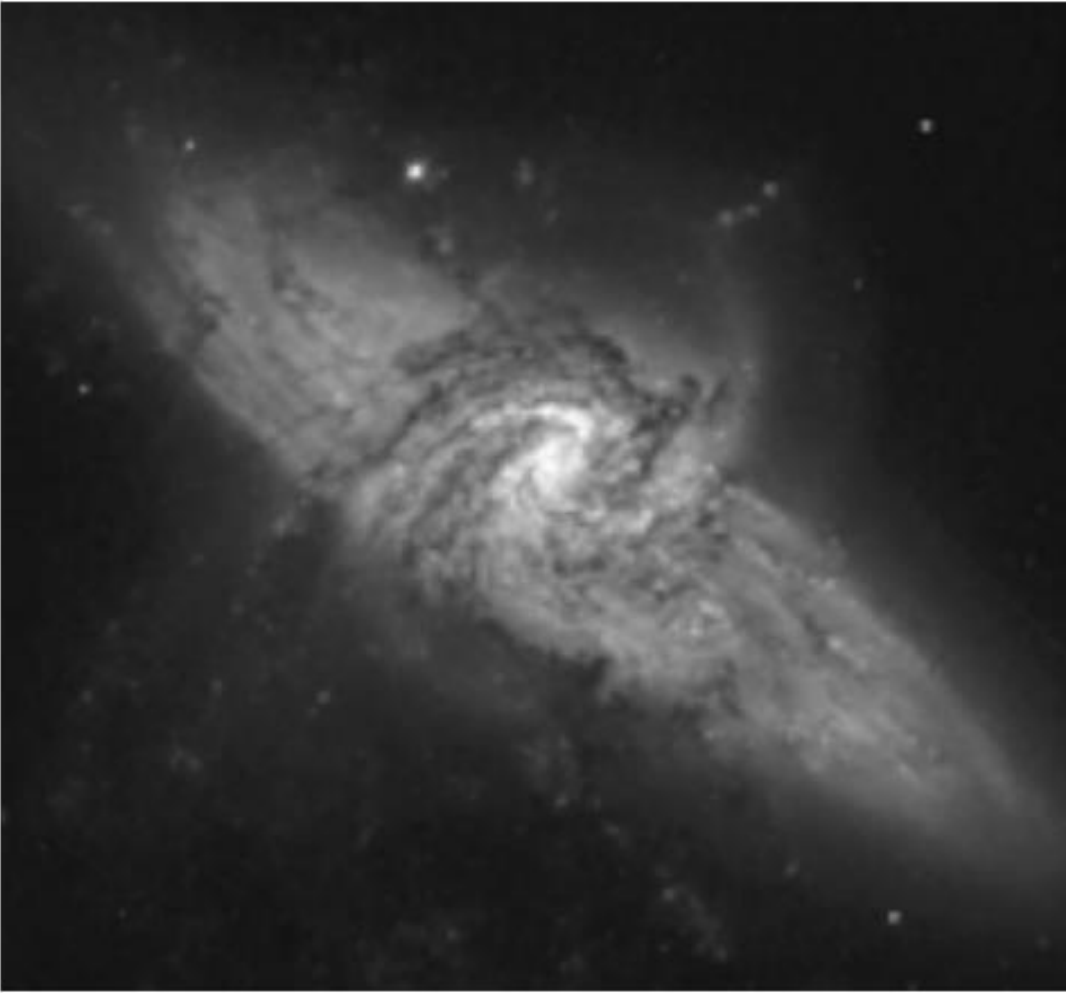
The standard deviation at any point in the average image is

$$\sigma_{\bar{g}(x,y)} = \frac{1}{\sqrt{K}} \sigma_{\eta(x,y)}$$

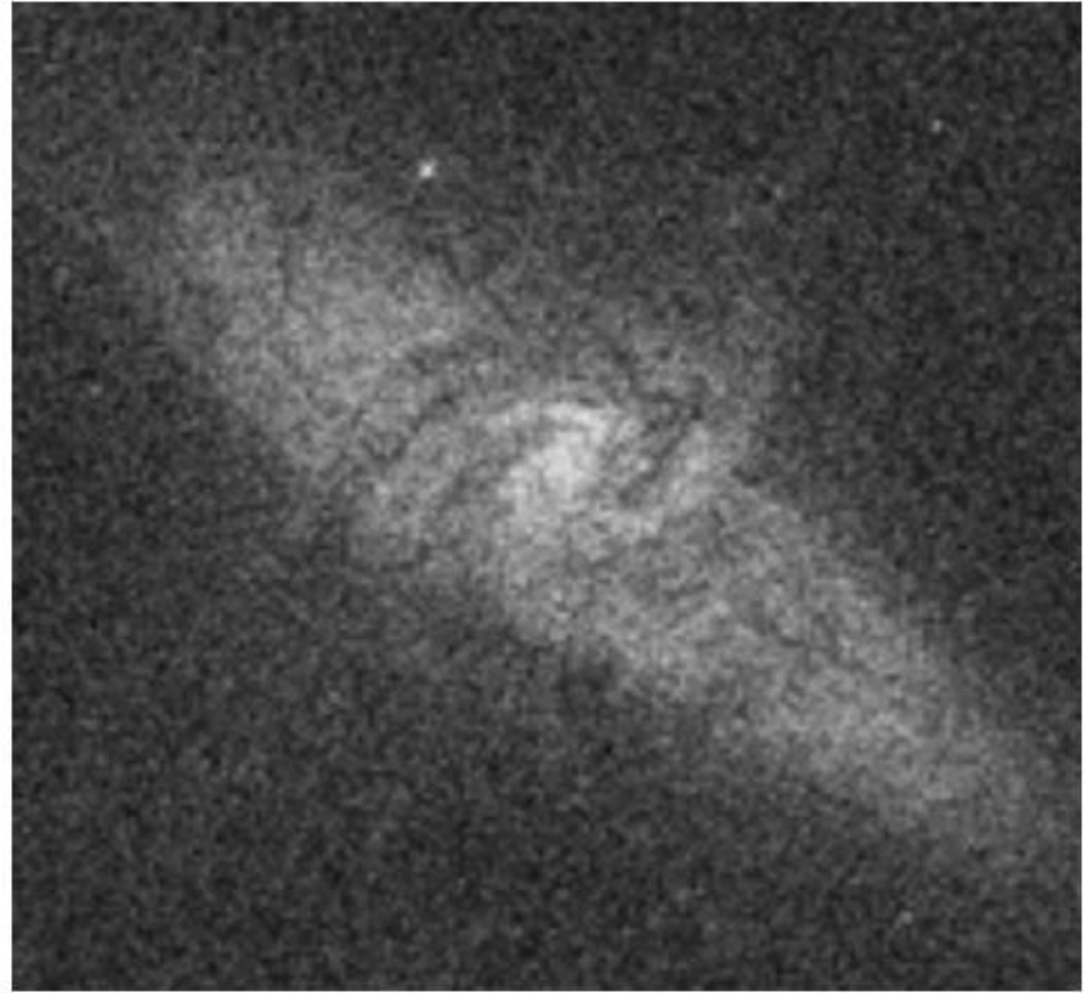
As K increases, the variability (noise) of the pixel values at each location (x, y) decreases.

$$E\{\bar{g}(x, y)\} = f(x, y)$$

this means that $\bar{g}(x, y)$ approaches $f(x, y)$ as the number of noisy images used in the averaging process increases.

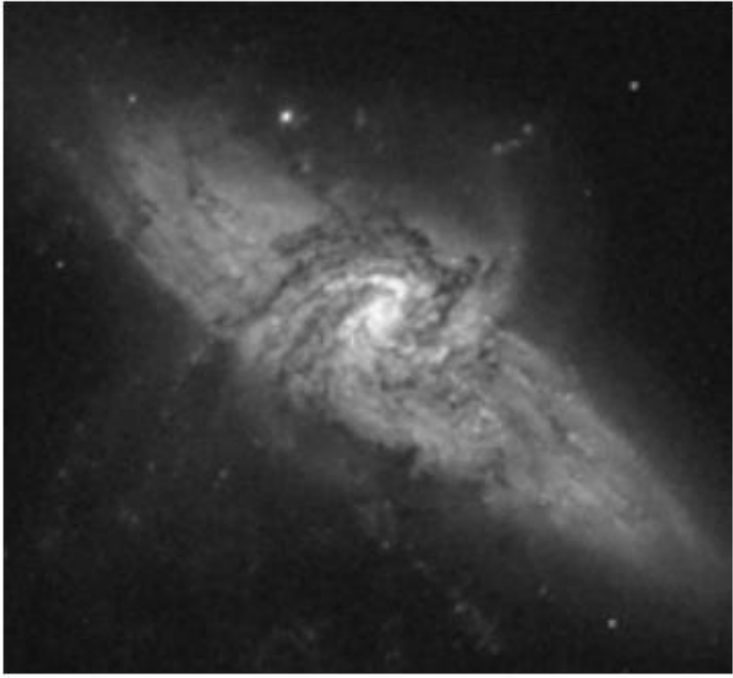


(a) Image of Galaxy Pair NGC 3314

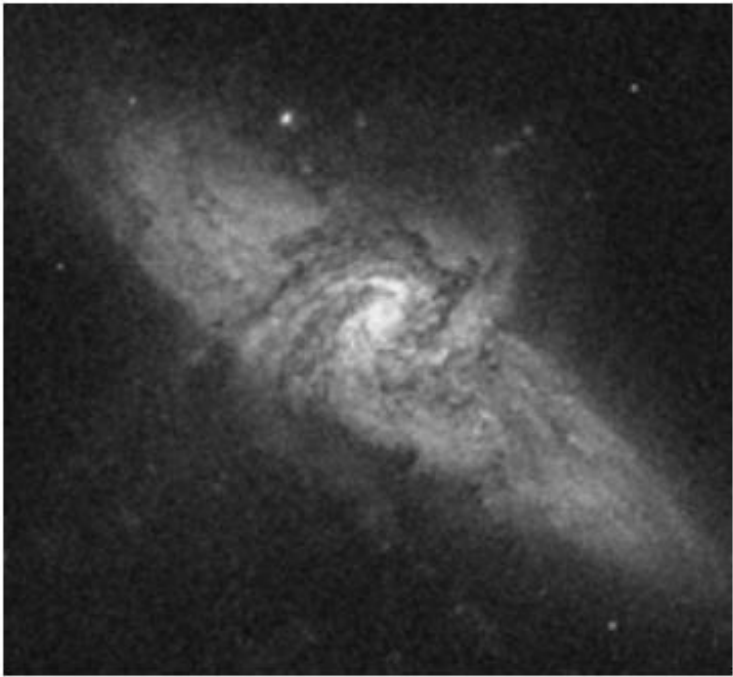


(b) Image corrupted by additive Gaussian noise with zero mean and a standard deviation of 64 gray levels.

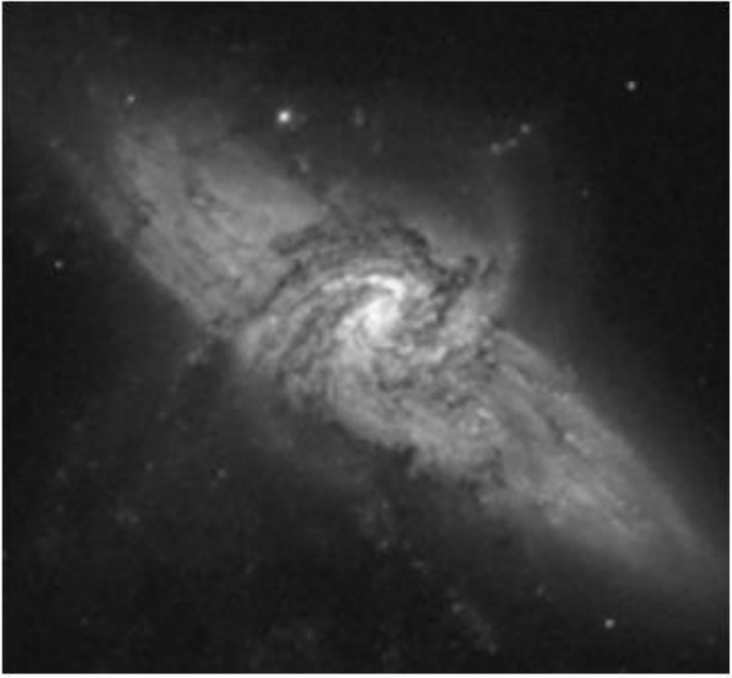
Results of
averaging $K=64$
noisy images.



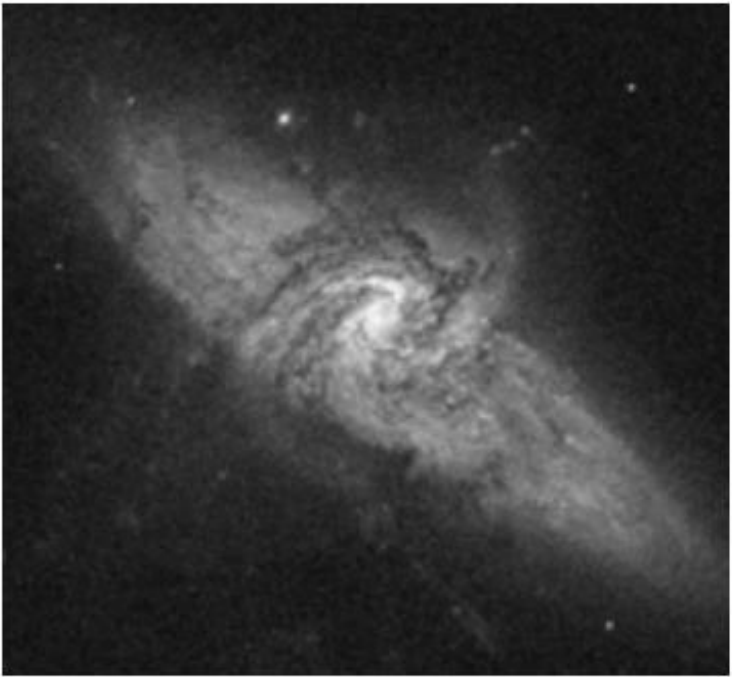
Results of
averaging $K=8$
noisy images.



Results of
averaging $K=128$
noisy images.



Results of
averaging $K=16$
noisy images.





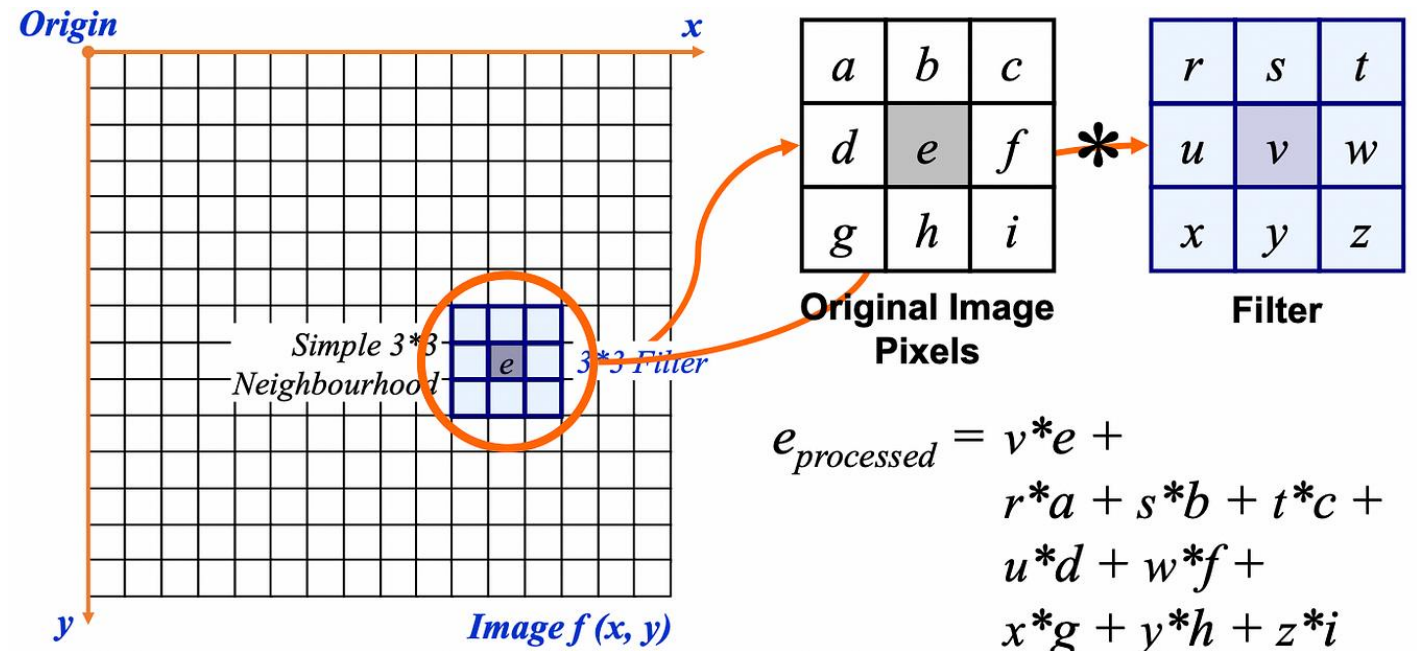
7. Spatial Filtering

Spatial Filtering technique is used directly on pixels of an image. Mask is usually considered to be added in size so that it has specific center pixel.

This mask is moved on the image such that the center of the mask traverses all image pixels.

The subimage is called a filter, mask, kernel, template, or window.

The values in a filter subimage are referred to as coefficients, rather than pixels.

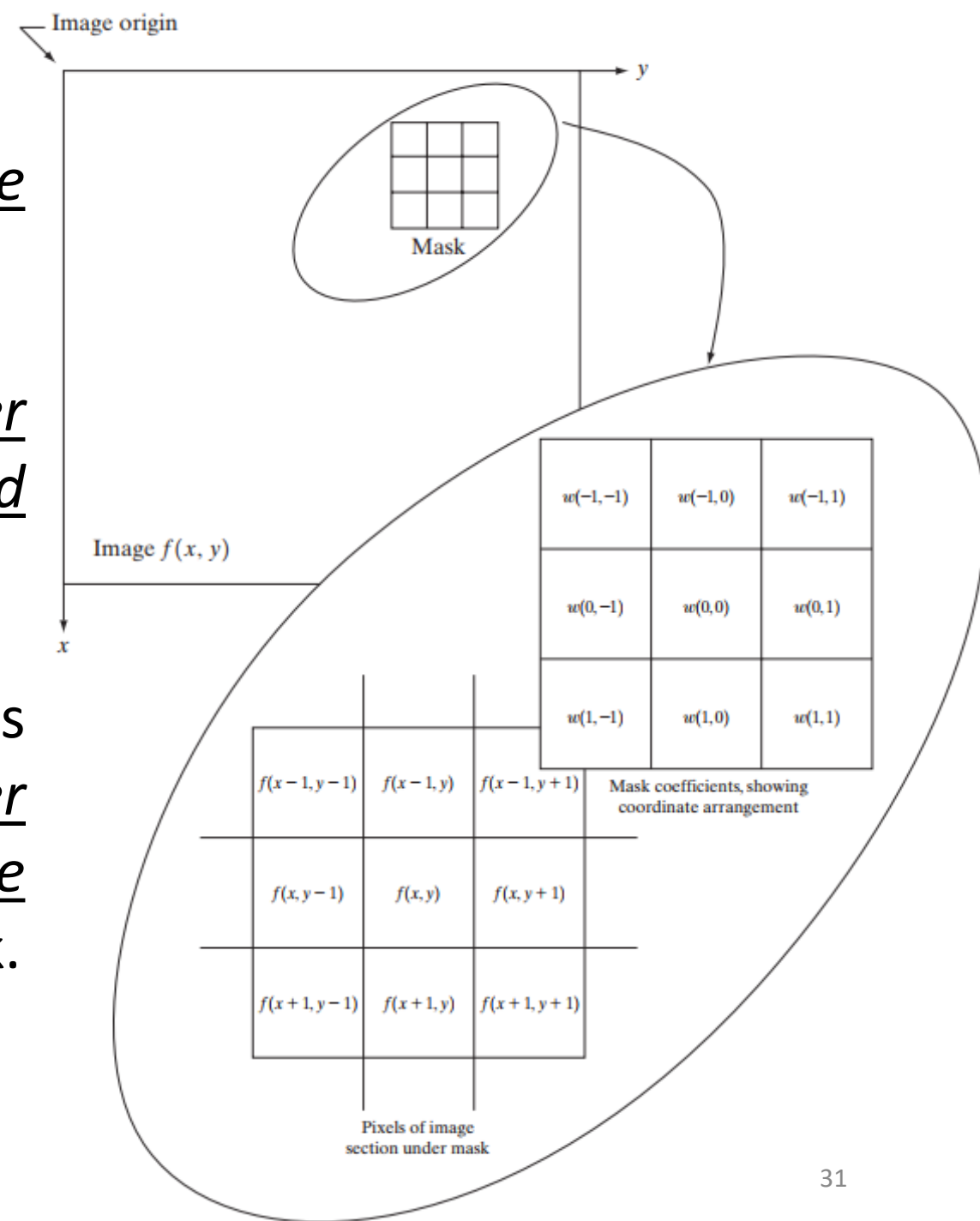


7.1 Mechanics of spatial filtering

The process consists simply of moving the filter mask from point to point in an image.

At each point (x, y) , the response of the filter at that point is calculated using a predefined relationship.

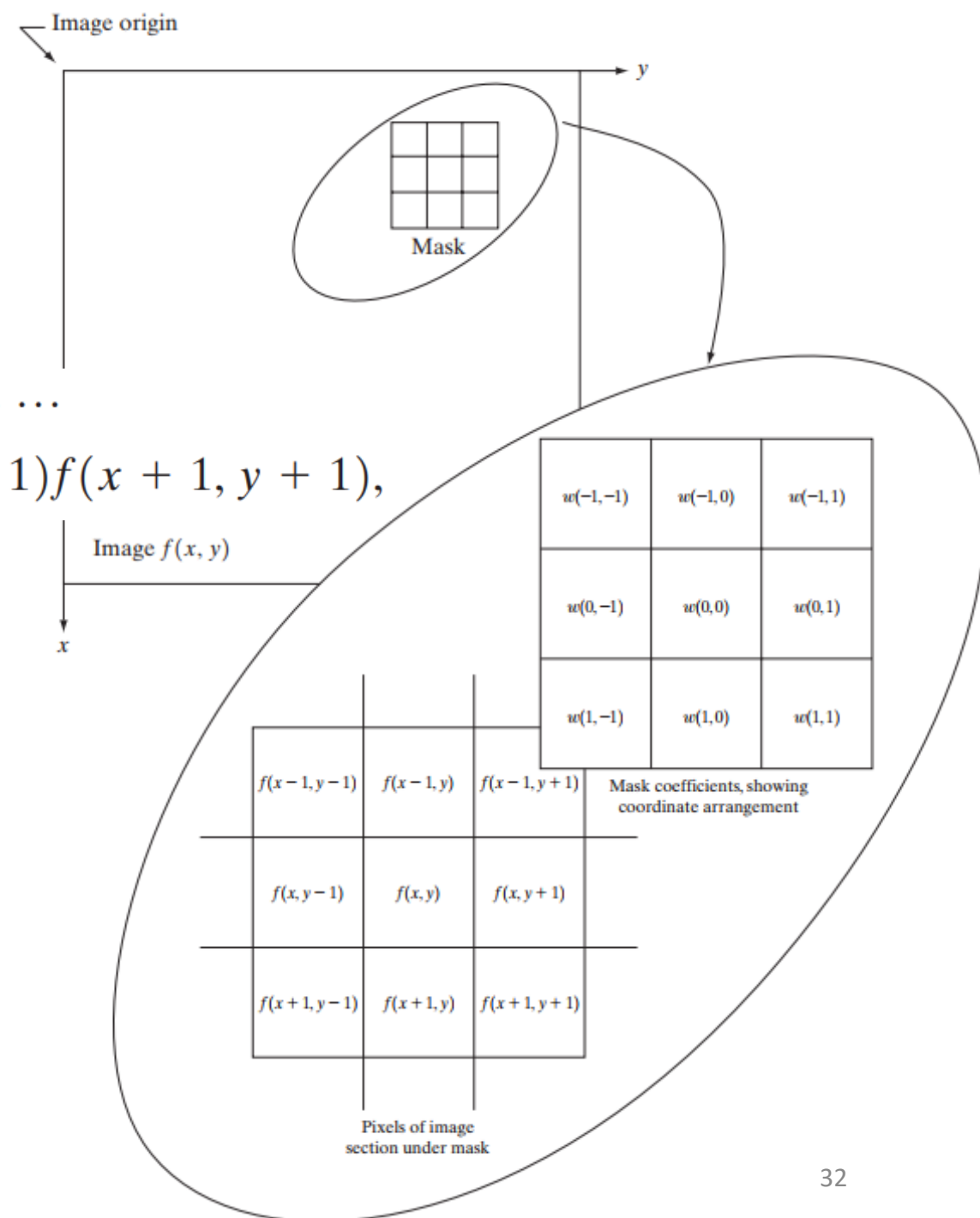
For linear spatial filtering, the response is given by a sum of products of the filter coefficients and the corresponding image pixels in the area spanned by the filter mask.



For the 3 X 3 mask, the result (or response), R , of linear filtering with the filter mask at a point (x, y) in the image is

$$R = w(-1, -1)f(x - 1, y - 1) + w(-1, 0)f(x - 1, y) + \cdots \\ + w(0, 0)f(x, y) + \cdots + w(1, 0)f(x + 1, y) + w(1, 1)f(x + 1, y + 1),$$

The coefficient $w(0, 0)$ coincides with image value $f(x, y)$, indicating that the mask is centered at (x, y) when the computation of the sum of products takes place.



7.2 Padding

In digital image processing, padding refers to the process of adding extra rows and/or columns around the border of an image.

This technique is commonly used to ensure that specific operations, such as filtering, convolution, or resizing, are applied consistently across all parts of the image, including its edges.

Zero Padding

Constant
Padding

Nearest
Neighbor
Padding

Reflect
Padding

Wrap Padding

7.2.1 Zero Padding

Zero padding is used to add extra rows and columns of zeros to the edges of an image.

This technique is also known as “border padding” or “border mode”.

The zeros are added to the borders of the image so that the size of the image is increased, but the original content remains unchanged.

0	2	5	7	3	10	9
11	1	4	6	8	2	0
0	12	10	9	7	4	5
1	9	7	8	13	11	0
5	10	14	6	2	1	1
7	6	11	3	13	8	4
3	9	6	12	7	10	5

(a) A 7-by-7 input image.

0	0	0	0	0	0	0	0	0	0	0
0	0	0	2	5	7	3	10	9	0	0
0	0	11	1	4	6	8	2	0	0	0
0	0	0	12	10	9	7	4	5	0	0
0	0	1	9	7	8	13	11	0	0	0
0	0	5	10	14	6	2	1	1	0	0
0	0	7	6	11	3	13	8	4	0	0
0	0	3	9	6	12	7	10	5	0	0
0	0	0	0	0	0	0	0	0	0	0

(b) Padding with zeros.

7.2.2 Constant Padding

This type of padding is suitable when the areas near the border of the image or data have a relatively constant or uniform pattern that we want to maintain.

0	2	5	7	3	10	9
11	1	4	6	8	2	0
0	12	10	9	7	4	5
1	9	7	8	13	11	0
5	10	14	6	2	1	1
7	6	11	3	13	8	4
3	9	6	12	7	10	5

(a) A 7-by-7 input image.

5	5	5	5	5	5	5	5	5	5	5
5	5	0	2	5	7	3	10	9	5	5
5	5	11	1	4	6	8	2	0	5	5
5	5	0	12	10	9	7	4	5	5	5
5	5	1	9	7	8	13	11	0	5	5
5	5	5	10	14	6	2	1	1	5	5
5	5	7	6	11	3	13	8	4	5	5
5	5	3	9	6	12	7	10	5	5	5
5	5	5	5	5	5	5	5	5	5	5

(c) Padding with a constant.

7.2.3 Nearest Neighbor Padding

0	2	5	7	3	10	9
11	1	4	6	8	2	0
0	12	10	9	7	4	5
1	9	7	8	13	11	0
5	10	14	6	2	1	1
7	6	11	3	13	8	4
3	9	6	12	7	10	5

(a) A 7-by-7 input image.

0	0	0	2	5	7	3	10	9	9	10
0	0	0	2	5	7	3	10	9	9	9
11	11	11	1	4	6	8	2	0	0	0
0	0	0	12	10	9	7	4	5	5	5
1	1	1	9	7	8	13	11	0	0	0
5	5	5	10	14	6	2	1	1	1	1
7	7	7	6	11	3	13	8	4	4	4
3	3	3	9	6	12	7	10	5	5	5
3	3	3	9	6	12	7	10	5	5	5

(d) Padding with nearest neighbor.

7.2.4 Reflect Padding

Reflect padding, also known as mirror padding or symmetric padding, is a technique used to create a reflected boundary around an image. In this technique, the pixels at the edges of the image are mirrored to create a boundary of reflected pixels.

0	2	5	7	3	10	9
11	1	4	6	8	2	0
0	12	10	9	7	4	5
1	9	7	8	13	11	0
5	10	14	6	2	1	1
7	6	11	3	13	8	4
3	9	6	12	7	10	5

(a) A 7-by-7 input image.

2	0	0	2	5	7	3	10	9	9	10
2	0	0	2	5	7	3	10	9	9	10
1	11	11	1	4	6	8	2	0	0	2
12	0	0	12	10	9	7	4	5	5	4
9	1	1	9	7	8	13	11	0	0	11
10	5	5	10	14	6	2	1	1	1	1
6	7	7	6	11	3	13	8	4	4	8
9	3	3	9	6	12	7	10	5	5	10
9	3	3	9	6	12	7	10	5	5	10

(e) Padding with reflect option.

7.2.5 Wrap Padding

Wrap padding, also known as cyclic padding or periodic padding, is a technique used to create a new image by wrapping the original image around its edges. In this technique, the pixels at the edges of the image are copied and appended to the opposite side of the image.

0	2	5	7	3	10	9
11	1	4	6	8	2	0
0	12	10	9	7	4	5
1	9	7	8	13	11	0
5	10	14	6	2	1	1
7	6	11	3	13	8	4
3	9	6	12	7	10	5

(a) A 7-by-7 input image.

5	10	3	9	6	12	7	10	5	3	9
9	10	0	2	5	7	3	10	9	0	2
0	2	11	1	4	6	8	2	0	11	1
5	4	0	12	10	9	7	4	5	0	12
0	11	1	9	7	8	13	11	0	1	9
1	1	5	10	14	6	2	1	1	5	10
4	8	7	6	11	3	13	8	4	7	6
5	10	3	9	6	12	7	10	5	3	9
9	10	0	2	5	7	3	10	9	0	2

(f) Padding with wrap option.